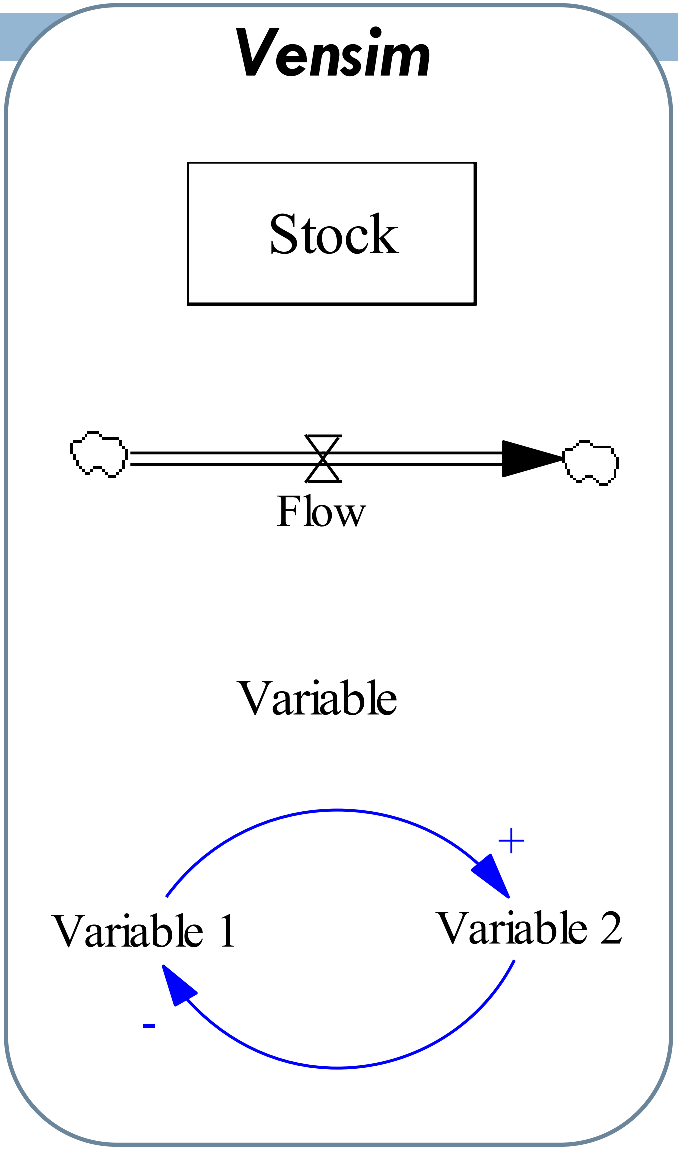
OPERATIONS RESEARCH AND SYSTEMS ANALYSIS

System dynamics - tutorial

Stock and flow diagram

- *Stock, level, box variable*
- *Flow*
- *Auxiliary variable, constant*
- *Causal link, arrow*



System representation (I)

- Stock and flow diagram:

- Integral:

$$Stock(T) = \int_{T_0}^{T} [Input(t) - Output(t)]\, dt + Stock(T_0)$$

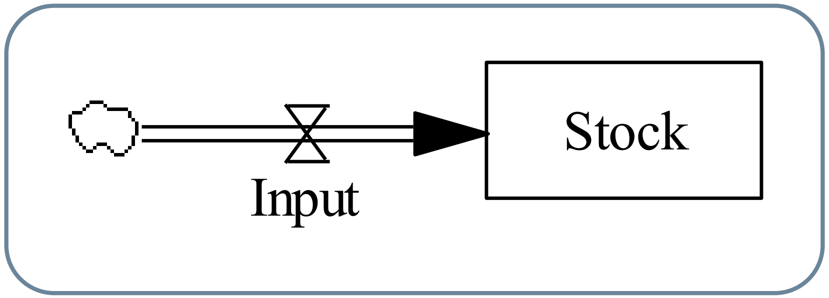
- Differential:

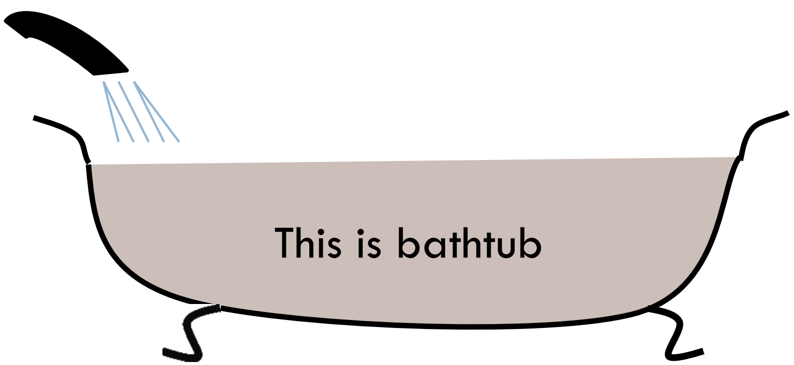
$$\frac{d(Stock)}{dt} = Input(t) - Output(t)$$

- Vensim:

$$Stock = INTEG\big(Input - Output, Stock(T_0)\big)$$

System representation (II)

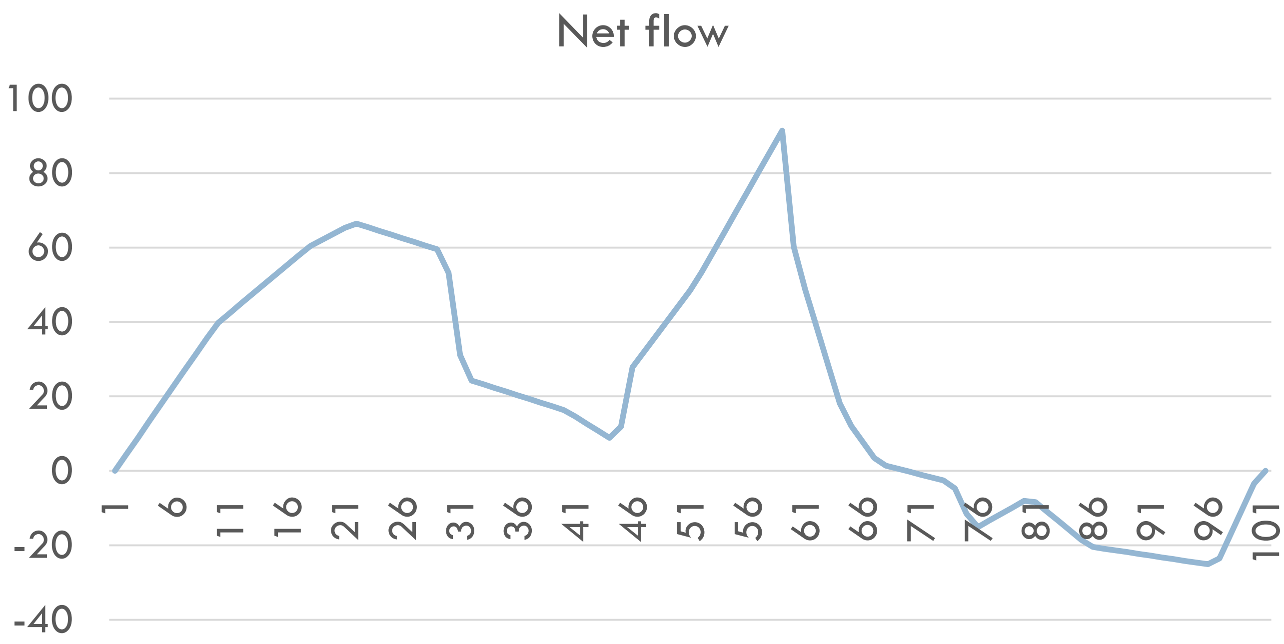




- Parameters in text form:
  - FINAL TIME =
  - INITIAL TIME =
  - TIME STEP =
  - Stock = INTEG (Input, 0)

Task 1

- The net flow (= inflow – outflow) has following course:



- Draw the course of the stock variable if the initial value of the stock is 0

Task 2

- Your initial balance is 10 000 EUR
- Your average annual earnings are 25 000 EUR
- Your average annual expenditures are 20 000 EUR
- The interest rate is 3%

- What will be your balance after 30 years.

- Build the model of that problem in Vensim PLE.

Infection

- On Island Total Population of 50,000 People, 10 Persons are Infected by Flu. Each Day, Every Person is 10 Times (in Average) in Contact with Other Person in the Way that Could Cause the Infection. When the Contact is Between Susceptible and Infected Persons, the Susceptible Person is Infected with 5% Probability.

- Start Vensim PLE Plus
- Build the model of infection
- Before the simulation, try to estimate how many people will be infected by flu after 100 days

- How would you extend the model?
  - Do it!

Thank you for your attention